Pharmaceuticals and Natural Products

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Pharmaceuticals and Natural Products (A07731)

Short Title: Pharmaceuticals & Nat Products
Faculty Science and Computing
Department Science
Cluster Pharmaceutical Science
Author CLENNON
Credits 5 **Level:** Intermediate

Description of Module / Aims

This module gives an insight into the structure and synthesis of some natural products and pharmaceuticals. A particular emphasis will be on the use of carbanions in the synthesis of such molecules.

Programmes

PHAR-0005	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	3 / 5 / M
PHAR-0005	BSc in Pharmaceutical Science (WD_SPHSC_D)	3 / 5 / M

Indicative Content

- The use of carbanions in the synthesis of a range of pharmaceuticals and natural products; synthetic pathways of these reactions along with mechanisms are studied in detail. Strategies involving the following reactions are included: α-halogenation, enolate alkylation, aldol, Claisen, Michael and Wittig reactions. Combinations of such reactions to generate cyclic products are discussed along with strategies for selectivity
- Structure of a range of pharmaceutical and natural products, including terpenes, steroids, sulphonamides and beta-lactam antibiotics
- Chemical aspects and the synthesis of peptides, the structure of proteins, nucleotides and DNA

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Differentiate and illustrate the structure of a range of pharmaceuticals and natural products.
2. Examine synthetic routes to suggested products.
3. Relate the mechanisms involved in representative reactions to synthesise such targets.
4. Use laboratory techniques to perform a multi-step organic synthesis.

Learning and Teaching Methods

- Lectures.
- Tutorials
- Laboratory-based practicals.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3
Continuous Assessment	50%	
Lab Report	40%	4
Tutorial/Problem Sheets	10%	1,2,3

Assessment Criteria

<40%: Little or no demonstration of an understanding of the fundamentals of pharmaceuticals and natural products. Lack of competence with regard to associated laboratory skills and practices.

40%-49%: Will be able to show a limited understanding of the fundamentals of pharmaceuticals and natural products. May have some difficulty with applying this theory. Will display some ability with regard to associated laboratory skills and practices.

50%-59%: As well as the above, the student will be to show an understanding of the complexities of pharmaceuticals and natural products. Will display reasonable competence with regard to associated laboratory skills and practices.

60%-69%: In addition, the student will demonstrate a more detailed knowledge of pharmaceuticals and natural products and display a very good level of competence and efficiency with regard to associated laboratory skills and practices.

70%-100%: All of the above to an excellent level. In addition, the learner will be able to demonstrate comprehensive knowledge and understanding of pharmaceuticals and natural products, as well as being able to predict general trends, evaluate and relate topics. Will display advanced competence and independence with regard to associated laboratory skills and practices.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	9	
Lab	27	
Independent Learning	75	

Essential Material

- "Mastering Chemistry." www.pearsonmylabandmastering.com/masteringchemistry
- Bruice, P.Y. _Organic Chemistry_. 8th Edition. New Jersey: Pearson, 2010.

Supplementary Material

- "The Virtual Textbook of Organic Chemistry." http://www2.chemistry.msu.edu/faculty/reusch/VirtTxtJml/intro1.htm

- Clayden, J., N. Greaves and S. Warren . _Organic Chemistry_. 2nd ed. Oxford: Oxford University Press, 2012.
- Lednicer, D. and L.A. Mitscher. _The Organic Chemistry of Drug Synthesis_. New York: Wiley, 1990.
- Mann, J., J. Harborne and R.S. Davidson. _Natural Products: Their Chemistry and Biological SignificancePublication info. Harlow : Longman Scientific & Technical, 1994._ . UK: Longman, 1994.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Chemistry Lab